Highlights

Thermal-Crosstalk-Aware Scheduling and Model Predictive Dynamic Voltage and Frequency Scaling for Spaceborne Multi-Chip AI Clusters in Deep Vacuum

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- Non-linear CTM network and EKF state estimation under deep orbital vacuum.
- LTV MPC-DVFS with soft thermal cap and explicit fallback damps crosstalk.
- Dynamic radiative power budgeting derived from Fourier-enthalpy balance.
- 3-orbit simulation exhibits zero 85 C violations and 78.7% gradient drop.
- Stratified LHS sensitivity ($n = 30$) shows zero violations in $\pm 20\%$ range.

Thermal-Crosstalk-Aware Scheduling and Model Predictive Dynamic Voltage and Frequency Scaling for Spaceborne Multi-Chip AI Clusters in Deep Vacuum

Dafnhê T. C. M. Barros[a,*], Flávio B. Braz da Silva[a]

[a]Graduate Program in Control and Automation Engineering (ProPECAut), Federal Institute of Espírito Santo (IFES), Rodovia BR-101 Norte, Km 58, Serra, 29166-064, Espírito Santo, Brazil

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ABSTRACT

High-density heterogeneous artificial intelligence (AI) computing clusters onboard small satellites in Low Earth Orbit (LEO) face severe electrothermal constraints. Gas-mediated convective heat transfer is neglected in the adopted high-vacuum model ($h \approx 0$), confining thermal rejection exclusively to internal solid conduction and non-linear Stefan-Boltzmann radiation to deep space ($q \propto T^4$). In multi-chip modules (MCM), heavy neural inference workloads induce severe lateral conductive coupling between adjacent silicon dies (*thermal crosstalk*). This paper investigates a model-based predictive thermal management strategy, denoted **TCAS-DVFS** (*Thermal-Crosstalk-Aware Scheduling and Model Predictive Dynamic Voltage and Frequency Scaling*), formulated from lumped thermodynamic principles for multi-die spaceborne accelerators. The proposed architecture integrates: (i) online state estimation at the simulated sampling rate via an Extended Kalman Filter (EKF) operating over a non-linear 5×5 Compact Thermal Model (CTM) with exact analytical subthreshold leakage and radiative Jacobians; (ii) a discrete linear time-varying Model Predictive Controller (LTV MPC-DVFS) that penalizes violations of a nominal 83 °C soft thermal cap via decoupled spatio-temporal slack variables, solved via a warm-started ADMM splitting iteration paired with an explicit shifted-tail power-projected fallback for non-convergent iterations; (iii) a physical radiative and conductive power-budget closure with transient enthalpy buffering derived from a synthetic analytical orbital radiative forcing model; and (iv) a topological task allocator relaxed over the semidefinite cone $\mathbb{S}_+^{17}$ via Semidefinite Programming (SDP). In a three-orbit numerical co-simulation of a 500-km circular LEO scenario (17,004.0 s, 34,009 time samples, 4 processing dies), TCAS-DVFS produced zero observed junction temperature violations above the adopted 85 °C simulation thermal limit (0 s versus 200.5 s for binary Thermal Throttling at 85/75 °C and 14.0 s for Space-PID), attenuated peak inter-die thermal gradients by 78.7% (reducing from 9.42 to 2.01 K, and mean gradients by 83.1% from 5.26 to 0.89 K), and maintained an aggregate throughput of 124.62 FPS with an energy-delay product proxy ($E_{t_{\text{frame}}}$) of 7.452 kJ·s/frame. Functional ablation indicates a dominant contribution from the predictive control formulation in damping gradients and bounding temperatures, whereas topological SDP scheduling matches the best discrete permutation among the $4! = 24$ candidates after hyperplane rounding (142/142 epochs) despite partial ADMM numerical convergence, providing limited net thermal benefit in compact symmetric arrangements ($N = 4$ dies). Descriptive parametric sensitivity was evaluated under a 30-sample stratified Latin Hypercube Sampling (LHS) campaign ($\pm 20\%$ perturbation across the four conductive and capacitive network parameters R_{ij}, R_r, c_i, and C_{rad}), yielding zero observed violations (0/30 samples > 85 °C), where die-to-radiator thermal resistance (R_r) exhibited the strongest monotonic association with $T_{\max}$ (Spearman $\rho = +0.995$). Indicative reliability proxy analyses demonstrate that electromigration yields no net life extension (relative Black damage ratios of 0.97 and 0.83), whereas cyclic thermal stress severity is substantially damped. The reported evidence is limited to numerical simulation of the stated lumped-parameter model and does not constitute hardware or flight qualification.

1. Introduction

1.1. Orbital Edge Computing and Spaceborne AI

Modern Earth Observation (EO) satellite missions, including wildfire monitoring, maritime surveillance, disaster response, and Synthetic Aperture Radar (SAR) imaging, produce massive data streams operating at multi-gigabit-per-second rates in Low Earth Orbit (LEO) (Denby and Lucia, 2020; Garrett et al., 2024; Chen et al., 2026a; Ortiz del Castillo et al., 2024). Traditionally, remote sensing platforms have relied on the conventional *store-and-forward* architecture, wherein raw payload data are captured, buffered in solid-state recorders, and downlinked during brief 8-to-12-minute ground-station passes per orbit. This approach incurs severe latency penalties, frequently exceeding 4 to 12 hours (Denby and Lucia, 2020; Garrett et al., 2024).

Edge artificial intelligence (AI) onboard small satellites mitigates this downlink bottleneck by processing, compressing, and triaging sensor data directly in orbit using commercial off-the-shelf (COTS) neural accelerators, multicore ARM CPUs, and embedded GPUs (Garrett et al., 2024; Sherwood et al., 2003). Because global satellite observations indicate that cloud cover obscures approximately 67%

*Corresponding author.
thumbnails/cas-email.jpeg dafnhe.barros@ifes.edu.br (Dafnhê T. C. M. Barros);
flavio.silva@ifes.edu.br (Flávio B. Braz da Silva)
ORCID(s): 0009-0004-9840-0257 (Dafnhê T. C. M. Barros);
0000-0002-4752-9742 (Flávio B. Braz da Silva)

of the Earth's surface on average (King et al., 2013), automated onboard cloud-detection models such as CloudScout on Φ-Sat-1 screen and filter out cloud-obstructed scenes (e.g., discarding images exceeding 70% cloudiness) prior to transmission, conserving scarce communication bandwidth for actionable mission products (Giuffrida et al., 2022).

1.2. Deep Vacuum Thermodynamics and Thermal Crosstalk

Unlike terrestrial data centers that leverage forced air convection ($h \approx 20$–100 W/(m^{2}K)) or pumped liquid cooling loops ($h \approx 1{,}000$–$10{,}000$ W/(m^{2}K)) (Bergman et al., 2011), small spacecraft operating in the high vacuum of LEO ($P < 10^{-4}$ Pa, free molecular flow with Knudsen number $Kn \gg 10$) (Gilmore, 2002) exhibit negligible convective gas exchange. Consequently, gas-mediated convective heat transfer is neglected in the adopted high-vacuum model ($h \approx 0$). Thermal rejection from energized silicon junctions is restricted to two coupled mechanisms (European Cooperation for Space Standardization, 2008; Gilmore, 2002):

1. **Internal solid conduction:** heat dissipates from active junctions across thermal interface materials (TIM) into a shared cold plate or metallic carrier (e.g., AlN or Al 6061-T6, $k \approx 170$–200 W/(m · K)) (European Cooperation for Space Standardization, 2008). Direct radiative exchange between facing die sidewalls is negligible ($< 10^{-3}$ W due to microscopic lateral areas $A_{lat} \approx 10^{-5}$ m^2 and small temperature differences).
2. **External radiation to deep space:** structural heat reaches outer passive radiators and is radiated to the cold cosmic background ($T_{space} \approx 2.73$ K) governed by the non-linear Stefan-Boltzmann law ($q_{rad} \propto \varepsilon\sigma T^4$) (Bergman et al., 2011; Pederson et al., 2025).

In compact small-satellite form factors (e.g., 6U–12U CubeSats), volume and mass constraints dictate packaging multiple System-on-Module (SoM) accelerators on a single carrier. Simultaneous deep neural inference workloads dissipate between 10 and 25 W per die (Reuther et al., 2022; Kang and Park, 2025), demanding 60 to 100 W from the electrical power system (EPS). Crucially, lateral conductive flux propagating through the carrier raises the base temperature of neighboring idle or lightly loaded dies by 10 to 18 K within tens of seconds (Chen et al., 2026b; Wang et al., 2026). This mutual conductive coupling, termed **spatial thermal crosstalk**, induces mutual thermal interference and severely degrades uncoordinated, decentralized governors.

1.3. Deficiencies of Terrestrial Thermal Governors

Conventional reactive algorithms, such as the Linux *ondemand* governor (Pallipadi and Starikovskiy, 2006) and classical Single-Input Single-Output (SISO) PID loops, fail in orbital clusters due to three structural factors (Brooks and Martonosi, 2001; Skadron et al., 2004; Hoffmann and Fröhlich, 2022):

1. **Large thermal inertia:** passive radiators ($C_{rad} \approx 1{,}300$ J/K) introduce dominant time constants of tens of minutes, causing purely reactive policies to act with critical delay.
2. **Limit-cycle saw-tooth oscillations:** binary frequency throttling upon crossing emergency alarm thresholds triggers repetitive temperature fluctuations ($\Delta T \approx 10$ K).
3. **Spatial gradient amplification:** uncoordinated DVFS across asymmetric workloads exacerbates instantaneous temperature differentials across adjacent dies, accelerating low-cycle thermomechanical shear fatigue in Ball Grid Array (BGA) solder interconnects (Manson, 1966).

1.4. Research Question and Contributions

The pioneering work of Chen et al. (2026b) established thermal-crosstalk-aware task scheduling in spaceborne AI clusters. Building on this domain, our work investigates the following research question:

> *Can a model-based coordinated MPC-DVFS policy reduce thermal-threshold exceedance and inter-die temperature gradients in a simulated four-die orbital AI cluster while maintaining competitive inference throughput relative to reactive thermal governors under identical modeled workload and environmental conditions?*

To address this question through model-based numerical simulation, this paper presents the following contributions and findings within the scope of the modeled multi-physics system:

1. **Non-linear State-Space Electrothermal Modeling:** we formulate a 5×5 compact thermal network (CTM) capturing conductive coupling matrices, linearized Stefan-Boltzmann radiator dynamics, and non-linear subthreshold leakage positive feedback, integrated with an Extended Kalman Filter (EKF) observer under stochastic sensor noise.
2. **Constrained Predictive Control (LTV MPC-DVFS):** we formulate a discrete linear time-varying Model Predictive Controller that solves a condensed quadratic programming problem via a warm-started ADMM-style splitting iteration with an explicit safe fallback policy for non-convergent updates, coordinating operating frequencies to track inference workloads, penalize violations of a nominal 83 °C soft thermal cap, and suppress spatial gradients.
3. **Orbital Radiative-Conductive Power Budget Closure:** we derive a coupled Fourier-conduction and radiator-enthalpy power budget driven by a synthetic analytical orbital radiative forcing model to establish a temperature-referenced power envelope across eclipse transitions.
4. **Numerical Sensitivity and Methodological Demarcation:** we execute a three-orbit numerical co-simulation, functional ablation, a descriptive 30-sample stratified Latin Hypercube Sampling (LHS)

Table 1
Methodological Demarcation of Study Elements

Methodological Category	Specific Elements in this Study
Input Physical Parameters	Conductive resistances R_{ij}, die-to-radiator resistance R_r, die capacitances c_i, radiator capacitance C_{rad}, optical emissivity ε, solar absorptivity α_s (Table 2)
Model Hypotheses & Structure	Lumped-parameter 5×5 CTM; gas conduction neglected ($h\approx0$); linearized radiative state coupling; planar symmetry
Synthetic Workload Inputs	Continuous activity factors $\alpha_i(t)$ and nominal requested frequencies $f_{nom,i}(t)$ across 4 mission workloads
Model Environmental Forcing	Direct solar flux $Q_{sol}(t)$, Earth albedo $Q_{alb}(t)$, and terrestrial infrared $Q_{IR}(t)$ across a 500-km circular orbit
Calculated State Variables	Physical silicon junction temperatures $T_i(t)$ $(i=1,\ldots,4)$ and radiator core temperature $T_{rad}(t)$
Estimated State Variables	EKF state estimates $\hat{T}_i(t)$ and $\hat{T}_{rad}(t)$ reconstructed from noisy sensor measurements ($\sigma_s=0.25$ K)
Actuation Control Variables	Operating clock frequencies $\mathbf{f}(t)=[f_1(t),\ldots,f_4(t)]^T\in[0.30,1.30]$ GHz
Direct Numerical Outputs	Maximum temperature $T_{\max}$, spatial gradient $\Delta T_{esp}(t)$, violation time $t_{>85}$, total energy E
Evaluated Computational Proxies	Aggregate throughput proxy FPS, energy-delay product proxy $E_{t_{\text{frame}}}$, relative Black electromigration damage ratio, thermal-history rainflow cycle distribution
Unmeasured Physical Quantities	Physical component service life, BGA viscoplastic solder strain, internal package microgradients, space radiation effects (TID/SEE), and flight qualification

sensitivity analysis across the four passive conductive and capacitive network parameters (R_{ij},R_r,c_i,C_{rad}), and physics-of-failure reliability proxy evaluations.

To ensure complete methodological transparency, Table 1 establishes the explicit demarcation between input assumptions, synthetic drivers, calculated simulation states, evaluated proxies, and physically unmeasured quantities.

The remainder of this article is organized as follows: Section 2 presents the orbital environmental physics and radiative balance. Section 3 derives the electrothermal state-space model. Section 4 develops the TCAS-DVFS control architecture. Section 5 details the co-simulation methodology. Section 6 presents comparative simulation results, ablation, and parametric sensitivity. Section 7 evaluates indicative reliability proxies. Section 8 discusses related work, Section 9 discusses practical implementation considerations and threats to validity, and Section 10 concludes the paper.

2. Orbital Environmental Physics and Heat Balance

2.1. External Orbital Radiative Fluxes

We consider a synthetic analytical orbital radiative forcing model for a circular low Earth orbit at altitude $h=500$ km ($R=R_E+h=6{,}871$ km, with $R_E=6{,}371$ km and $\mu_E=398{,}600.4418$ km^3/s^2), yielding an analytical orbital period $T_{orb}=5{,}668.14$ s (≈94.47 min). Rather than propagating numerical two-line element (TLE) ephemerides, the external environmental flux is synthesized from the analytical orbital phase $\varphi=\mathrm{mod}(t,T_{orb})/T_{orb}$ under explicit geometric simplifications: a fixed solar beta angle $\beta=0^\circ$ (maximizing eclipse duration and eclipse-to-sunlight thermal shock), constant attitude incidence factors ($\cos30^\circ\cos15^\circ$), a constant Earth-to-radiator view factor

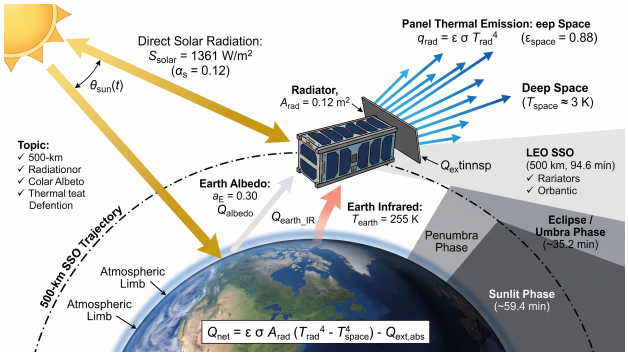


Figure 1: Synthetic 500 km circular LEO orbital geometry (Sun-Synchronous Orbit inclination $i\approx97.4^\circ$, solar beta angle $\beta=0^\circ$): direct solar flux, Earth albedo, terrestrial infrared emission, eclipse cone, and deep space radiator rejection. Conceptual rendering.

$F=0.42$, and a sinusoidal parameterization of effective terrestrial infrared temperature. With illuminated orbital fraction $\varphi_{sol}=1-\arcsin(R_E/R)/\pi$, orbital solar noon at $\varphi_m=\varphi_{sol}/2$, and smooth eclipse transition operator $\mathbb{I}_{sun}(t)=[1+e^{120(\varphi-\varphi_{sol})}]^{-1}$, the absorbed radiative fluxes on a radiator of area A_{rad} (Fig. 1) are given by:

$$\cos\theta_{ef}(t)=\max\left[0,\ \cos\left(\frac{\pi(\varphi-\varphi_m)}{2\varphi_m}\right)\cos30^\circ\cos15^\circ\right], \tag{1}$$

$$Q_{sol}(t)=\mathbb{I}_{sun}(t)\,S_{solar}\,\alpha_s A_{rad}\cos\theta_{ef}(t), \tag{2}$$

$$Q_{alb}(t)=\mathbb{I}_{sun}(t)\,S_{solar}\,a_E\,\alpha_s A_{rad}F\cos\theta_{ef}(t), \tag{3}$$

$$Q_{IR}(t)=\varepsilon\sigma A_{rad}F\,T_E^4(t), \tag{4}$$

$$T_E(t)=255-15\sin^2\left(\pi\frac{\varphi-\varphi_m}{\varphi_{sol}}\right), \tag{5}$$

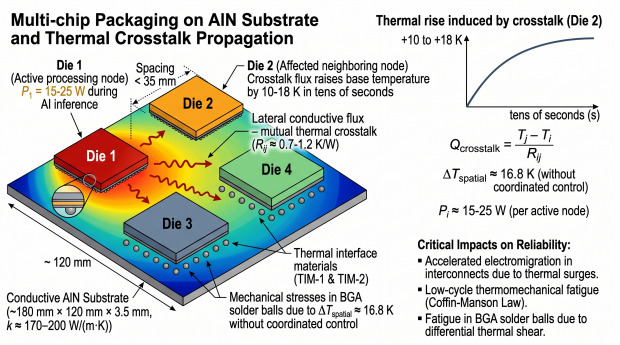


Figure 2: Solid conductive coupling and lateral thermal crosstalk in a multi-die carrier: heat dissipation in one node elevates base temperatures of adjacent dies and induces thermomechanical stress in BGA joints. Conceptual rendering.

where $S_{solar} = 1,361$ W/m^2 is the solar constant, $a_E = 0.30$ is the mean Earth albedo, $F = 0.42$ is the Earth-to-plate radiation view factor, $\alpha_s = 0.12$ is solar absorptivity, $\varepsilon = 0.88$ is infrared emissivity, and $\sigma = 5.67 \times 10^{-8}$ W/(m^{2}K^4) is the Stefan-Boltzmann constant. Earth albedo is modeled proportionally to the illuminated visible fraction via $\cos\theta_{ef}$, a standard conservative approximation in spacecraft thermal engineering (European Cooperation for Space Standardization, 2008; Gilmore, 2002). The total external flux $Q_{ext}(t) = Q_{sol}(t) + Q_{alb}(t) + Q_{IR}(t)$ varies between 8.34 W (in eclipse) and 29.09 W (solar noon), with an orbital mean of 16.84 W.

2.2. Passive Radiator Lumped Dynamics

The thermal energy conservation equation for the radiator panel of mass m_{rad} and specific heat $c_{p,rad}$ ($C_{rad} = m_{rad}c_{p,rad} = 1,299.2$ J/K) is governed by:

$$C_{rad}\frac{dT_{rad}}{dt} = \sum_{i=1}^{N}\frac{T_i - T_{rad}}{R_r} + Q_{ext}(t) - \varepsilon\sigma A_{rad}\left(T_{rad}^4 - T_{space}^4\right), \quad (6)$$

where $R_r = 2.10$ K/W represents the equivalent solid conductive thermal resistance between each die node and the radiator core.

3. State-Space Electrothermal Modeling

3.1. Silicon Power Dissipation and Leakage Feedback

Each silicon die (Fig. 2) dissipates electrical power partitioned into dynamic switching and subthreshold leakage components:

$$V_i = \mathrm{clip}(V_{min} + \beta_v(f_i - f_{min}), V_{min}, V_{max}), \quad (7)$$

$$P_{dyn,i} = \alpha_i C_{eff} V_i^2 f_i, \quad (8)$$

$$P_{leak,i} = P_{ref}\left(\frac{V_i}{V_0}\right)^{1.5}\left(\frac{T_i}{T_0}\right)^2 \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_0} - \frac{1}{T_i}\right)\right], \quad (9)$$

where total power is $P_i = P_{dyn,i} + P_{leak,i}$, with α_i denoting active workload factor, $C_{eff} = 24.5$ nF the adopted effective switching capacitance, and $\beta_v = (V_{max} - V_{min})/(f_{max} - f_{min})$. Leakage power $P_{leak,i}$ exhibits a quadratic-exponential dependence on absolute junction temperature T_i, introducing non-linear positive electrothermal feedback ($\partial P_{leak}/\partial T_i > 0$) and a risk of thermal runaway in the absence of active intervention. Rather than representing an experimental calibration of a specific hardware datasheet, the parameters $C_{eff} = 24.5$ nF, $P_{ref} = 0.80$ W (at $T_0 = 300$ K, $V_0 = 0.85$ V), and $E_a = 0.20$ eV are adopted as synthetic modeling assumptions scaled to match typical 10–25 W embedded accelerator power envelopes (Reuther et al., 2022; Kang and Park, 2025) (Table 2). Equation (7) is restricted to a nominal modeling domain ceiling of 125 °C; simulated trajectories exceeding 125 °C represent out-of-domain numerical extrapolations and are explicitly tagged (†).

3.2. Compact Thermal Network (5×5 CTM)

The planar thermal coupling across an array of $N = 4$ SoM nodes mounted on a shared AlN/Al carrier is modeled by a lumped-parameter compact thermal network:

$$c_i\frac{dT_i}{dt} = P_i(t) - \sum_{j\neq i}\frac{T_i - T_j}{R_{ij}} - \frac{T_i - T_{rad}}{R_r}, \quad (10)$$

where $c_i = 18.5$ J/K denotes the *effective local module node thermal capacitance*. Because a bare silicon die (~ 100 mm$^2 \times 0.5$ mm) possesses a heat capacity below 0.2 J/K, $c_i = 18.5$ J/K lumps the silicon junction together with its local metallic heat spreader/lid (e.g., ~ 20 g Cu/AlSiC), package substrate, TIM layer, and immediate carrier footprint beneath each module. In compact matrix notation:

$$\mathbf{C}\dot{\mathbf{T}} = \mathbf{P} - (\mathbf{G}_{crosstalk} + \mathbf{G}_{rad})\mathbf{T} + \mathbf{G}_{rad}T_{rad}\mathbf{1}_N, \quad (11)$$

where $\mathbf{C} = \mathrm{diag}(c_1, \ldots, c_N)$, $\mathbf{G}_{rad} = R_r^{-1}\mathbf{I}_N$, and $\mathbf{G}_{crosstalk} \in \mathbb{R}^{N\times N}$ is the symmetric conductive Laplacian matrix:

$$[\mathbf{G}_{crosstalk}]_{ij} = \begin{cases} -1/R_{ij}, & i \neq j, \\ \sum_{k\neq i} 1/R_{ik}, & i = j. \end{cases} \quad (12)$$

Modal decomposition of the die subsystem yields fast thermal time constants of 5.8, 7.5, 7.7, and 38.8 s, contrasting with the slow radiator time constant (1.5–2.1 ks). This pronounced two-time-scale separation provides thermodynamic justification for treating radiator temperature as a quasi-stationary state within the short-horizon predictive controller.

4. Integrated TCAS-DVFS Control Architecture

The closed-loop control architecture (Fig. 3) operates on a multi-rate scheme: continuous physical plant integration and EKF state filtering operate at $\Delta t_{\mathrm{plant}} = 0.5$ s, while model predictive control updates execute every $k_{sub} = 4$ steps ($\Delta t_{\mathrm{MPC}} = 2.0$ s), and topological task allocation updates occur at $\Delta t_{sched} = 120$ s.

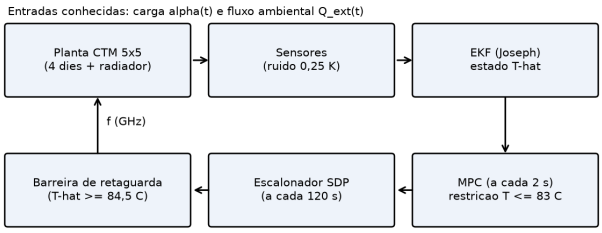


Figure 3: Closed-loop architecture of the TCAS-DVFS framework: non-linear 5×5 CTM plant, EKF state observer, orbital radiative power budgeter, topological SDP scheduler, and constrained discrete LTV MPC-DVFS controller.

4.1. State Estimation via Extended Kalman Filter

The extended state vector $\mathbf{x} = [T_1, T_2, T_3, T_4, T_{rad}]^T \in \mathbb{R}^5$ is measured under zero-mean additive white Gaussian sensor noise: $\mathbf{y} = \mathbf{x} + \mathbf{v}$, $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \sigma_s^2 \mathbf{I}_5)$, with standard deviation $\sigma_s = 0.25$ K (Zhang and Srivastava, 2011). State propagation and covariance updates employ the stabilized Joseph form (Simon, 2006):

$$\hat{\mathbf{x}}^- = \hat{\mathbf{x}}_{k-1} + \Delta t_{\text{plant}}\, \mathbf{f}(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{k-1}), \tag{13}$$

$$\Phi = \mathbf{I}_5 + \Delta t_{\text{plant}} \mathbf{F} + \tfrac{1}{2}(\Delta t_{\text{plant}} \mathbf{F})^2, \tag{14}$$

$$\mathbf{P}^- = \Phi \mathbf{P} \Phi^T + \mathbf{Q} \Delta t_{\text{plant}}, \tag{15}$$

$$\mathbf{K} = \mathbf{P}^-(\mathbf{P}^- + \mathbf{R})^{-1}, \quad \hat{\mathbf{x}} = \hat{\mathbf{x}}^- + \mathbf{K}(\mathbf{y} - \hat{\mathbf{x}}^-), \tag{16}$$

$$\mathbf{P} = (\mathbf{I}_5 - \mathbf{K})\mathbf{P}^-(\mathbf{I}_5 - \mathbf{K})^T + \mathbf{K}\mathbf{R}\mathbf{K}^T, \tag{17}$$

where process noise covariance is $\mathbf{Q} = \text{diag}(0.02\,\mathbf{I}_4, 0.05)$, measurement noise covariance is $\mathbf{R} = \sigma_s^2 \mathbf{I}_5$, initial error covariance is $\mathbf{P}_0 = \text{diag}(\mathbf{I}_4, 2)$, and analytical Jacobian $\mathbf{F} = \partial \mathbf{f} / \partial \mathbf{x}$ includes both the linearized Stefan-Boltzmann radiator flux ($4\varepsilon\sigma A_{rad} \hat{T}_{rad}^3 / C_{rad}$) and the analytical sub-threshold leakage derivative $\partial P_{leak,i} / \partial T_i = P_{leak,i}(2/T_i + E_a/(k_B T_i^2))$. In closed-loop co-simulation, the observer is full-rank observable ($\text{rank}(\mathscr{O}) = 5$) and achieves an average state tracking RMSE of 0.107 K (57.1% noise attenuation relative to $\sigma_s = 0.25$ K).

4.2. Physical Radiative and Conductive Power Budget Closure

Using the synthetic analytical orbital radiative forcing model over a macroscopic thermal lookahead window ($\tau_{horiz} = 240.0$ s) with predicted moving-average environmental flux $\bar{Q}_{ext}(k)$ and net radiative rejection $Q_{rad,liq}(k) = \varepsilon\sigma A_{rad}(\hat{T}_{rad}^4 - T_{space}^4) - \bar{Q}_{ext}(k)$, we derive a closed-form physical power budget $P_{bud}(k)$ combining Fourier conduction and radiator enthalpy buffering. Given minimum possible total die power $P_{min,cores}(k) = \sum_{i=1}^N P_i(\hat{T}_i, f_{min}, \alpha_i)$, the quasi-steady radiator ceiling and conductive RC time constant are:

$$T_{rad,max} = T_{cap} - \frac{P_{min,cores}(k)\, R_r}{N}, \quad \tau_{rad} = \frac{C_{rad}\, R_r}{N}. \tag{18}$$

The transient enthalpy buffer rate $P_{buffer}(k)$ and coupled Fourier conduction limit $P_{cond,max}(k)$ over τ_{horiz} are:

$$P_{buffer}(k) = \frac{C_{rad} \max(0, T_{rad,max} - \hat{T}_{rad})}{\tau_{horiz}} \left(1 - e^{-\tau_{horiz}/\tau_{rad}}\right), \tag{19}$$

$$P_{cond,inst}(k) = \frac{N \max(0, T_{cap} - \hat{T}_{rad})}{R_r}, \tag{20}$$

$$P_{cond,max}(k) = \min\left(P_{cond,inst}(k),\ \frac{P_{cond,inst}(k) + (\tau_{horiz}/\tau_{rad}) Q_{rad,li}}{1 + \tau_{horiz}/\tau_{rad}}\right. \tag{21}$$

$$P_{bud}(k) = \min\big(P_{cond,max}(k),\ Q_{rad,liq}(k) + P_{buffer}(k)\big). \tag{22}$$

If $P_{bud}(k) < P_{min,cores}(k)$, the state is explicitly flagged as physically infeasible (`physically_infeasible`) without artificial floor masking, triggering deterministic fallback to $\mathbf{f}(k) = f_{min} \mathbf{1}_N$.

4.3. Topological Task Scheduler (TCAS)

At task allocation epochs ($\Delta t_{sched} = 120$ s), $M = 4$ incoming neural inference workloads are assigned to the $N = 4$ dies to minimize conductive crosstalk weighted by the centered steady-state impedance matrix $\blacksquare = \blacksquare(\mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T)\blacksquare$, where $\blacksquare = (\mathbf{G}_{crosstalk} + \mathbf{G}_{rad})^{-1}$:

$$\min_{\mathbf{x} \in \Pi_N} \mathbf{x}^T \left[(\hat{\mathbf{p}}\hat{\mathbf{p}}^T) \otimes \blacksquare - \mu_{mig} \text{diag}(\mathbf{x}_{prev})\right] \mathbf{x}, \tag{23}$$

subject to bijective task assignment constraints $\sum_{i=1}^N X_{pi} = 1$ ($\forall p \in \{1, \ldots, N\}$) and $\sum_{p=1}^N X_{pi} = 1$ ($\forall i \in \{1, \ldots, N\}$), which can be written in compact linear form as $\mathbf{a}_k^T \mathbf{x} = 1$ for $k = 1, \ldots, 2N$, where $\mathbf{x} = \text{vec}(\mathbf{X}) \in \{0,1\}^n$ ($n = N^2 = 16$). The doubly-nonnegative semidefinite relaxation (Boyd and Vandenberghe, 2004) lifts $\mathbf{x}$ into $\mathbf{Y} \in \mathbb{S}_+^d \cap \mathbb{R}_+^{d \times d}$ ($d = n + 1 = 17$, indexed on $0 \le i, j \le n$):

$$\mathbf{Y} = \begin{bmatrix} 1 & \mathbf{x}^T \\ \mathbf{x} & \mathbf{X}_{lift} \end{bmatrix} \in \mathbb{S}_+^{17} \cap \mathbb{R}_+^{17 \times 17}, \quad \mathbf{E}\,\text{vec}(\mathbf{Y}) = \mathbf{e}. \tag{24}$$

To ensure complete mathematical reproducibility, the linear constraint system $\mathbf{E}\,\text{vec}(\mathbf{Y}) = \mathbf{e}$ ($\mathbf{E} \in \mathbb{R}^{289 \times 289}$) is assembled from four explicit row blocks:

(i) *Homogenization anchor (1 row):* $Y_{0,0} = 1$;

(ii) *Binary idempotence lifting ($n = 16$ rows):* $Y_{i,i} - Y_{0,i} = 0$ for $i = 1, \ldots, 16$, enforcing $\text{diag}(\mathbf{X}_{lift}) = \mathbf{x}$ (corresponding to $x_i^2 = x_i$);

(iii) *Explicit matrix symmetry ($d(d-1)/2 = 136$ rows):* $Y_{i,j} - Y_{j,i} = 0$ for all $0 \le i < j \le 16$;

(iv) *Column-lifted assignment constraints ($2N \times d = 8 \times 17 = 136$ rows):* for each of the $2N = 8$ row/column assignment indicator vectors $\mathbf{a}_k \in \{0,1\}^{16}$ and each matrix column $l \in \{0, \ldots, 16\}$, we enforce $\sum_{j=1}^{16} a_{k,j} Y_{j,l} - Y_{0,l} = 0$, obtained by multiplying $(\mathbf{a}_k^T \mathbf{x} - 1) = 0$ by each lifted coordinate $y_l \in \{1, x_1, \ldots, x_{16}\}$. Setting $l = 0$ (where $Y_{0,0} = 1$ and $Y_{j,0} = x_j$) directly enforces $\mathbf{a}_k^T \mathbf{x} = 1$, preserving exact primal assignment feasibility on the first row/column of $\mathbf{Y}$, while $l \ge 1$ tightens the off-diagonal blocks of $\mathbf{X}_{lift}$.

Summing the four blocks yields $1+16+136+136=289$ rows. Because the $2N=8$ bipartite assignment equations possess one classical linear dependency ($\sum_p\sum_i X_{pi}=\sum_i\sum_p X_{pi}$), its lifting across 17 columns combined with the symmetry and diagonal equations produces 38 exact algebraic redundancies, leaving $\mathrm{rank}(\mathbf{E})=251$ ($\sigma_{251}=0.3480$, whereas $\sigma_{252}=1.50\times10^{-14}$). The affine projection step $\mathrm{vec}(\mathbf{Y})=\mathbf{v}_{tg}-\mathbf{E}^{+}(\mathbf{E}\mathbf{v}_{tg}-\mathbf{e})$ inside the two-block consensus ADMM is therefore computed using a rank-regularized Moore-Penrose pseudoinverse ($\mathrm{rcond}=10^{-10}$), which discards the 38 algebraic-zero singular values and bounds $\|\mathbf{E}^{+}\|_2=2.87$. Candidate discrete allocations are extracted via randomized hyperplane rounding (24 draws plus first-column projection). In interpreting the scheduler validation results, four metrics must be strictly distinguished: (1) *ADMM numerical convergence* within the 120-iteration cap, which reaches 42.0% under warm start (mean 94.8 iterations, $r_{pri}\le 9.27\times10^{-3}$, $r_{dua}\le 3.39\times10^{-2}$) and 62.0% under cold start across 50 benchmark instances; (2) *relaxed cone-affine feasibility*, where $\|\mathbf{E}\,\mathrm{vec}(\mathbf{Y})-\mathbf{e}\|_\infty\le 5.25\times10^{-15}$, $\lambda_{\min}(\mathbf{Z}_1)\ge -1.08\times10^{-15}$, and $\min(\mathbf{Z}_2)=0$; (3) *post-rounding objective quality*; and (4) *empirical match with exhaustive permutation search* (hit = 100.0% across all 142/142 scheduling calls in the 3-orbit campaign). Because $N=4$ dies yield only $4!=24$ discrete permutations, this 100% empirical hit rate reflects the small discrete search space recovered by hyperplane rounding rather than full ADMM convergence or a certified dual optimality gap.

4.4. Discrete Linear Time-Varying Model Predictive Control (LTV MPC-DVFS)

To coordinate operating frequencies $\mathbf{f}(k)=[f_1,\dots,f_N]^T$ online at each control interval ($\Delta t_{\mathrm{MPC}}=2.0$ s), the control problem is formulated as a discrete Linear Time-Varying Model Predictive Controller over a control horizon $H_c=8$ steps (16.0 s) and prediction horizon $H_p=30$ steps (60.0 s) with decoupled spatio-temporal slack variables $\mathbf{s}\in\mathbb{R}_{+}^{NH_p}$ (120 slacks, 32 control frequencies, 152 decision variables in total):

$$\begin{aligned}\min_{\mathbf{f},\mathbf{s}}\sum_{j=1}^{H_c}\Big[&w\|\mathbf{f}(k+j-1)-\mathbf{f}_{nom}(k+j-1)\|^2\\&+R_w\|\mathbf{D}\mathbf{f}(k+j-1)\|^2\Big]\\&+\rho\sum_{j=1}^{H_p}\|\tilde{\mathbf{T}}_{grad}(k+j)\|^2+\lambda\|\mathbf{s}\|_2^2,\end{aligned}\tag{25}$$

subject to physical and operational constraints:

$$f_{min}\le f_i(k+j)\le f_{max},\quad \forall i,j,\tag{26}$$

$$T_i(k+j|k)\le T_{cap}+s_{i,j},\quad s_{i,j}\ge 0,\tag{27}$$

$$\sum_{i=1}^{N}P_i(k+j|k)\le P_{bud}(k+j),\quad \forall j,\tag{28}$$

where $w=1.0$ penalizes deviation from nominal inference request $\mathbf{f}_{nom}$, $R_w=0.2$ regularizes frequency change rate via the first-difference operator $\mathbf{D}$, $\rho=0.02$ penalizes spatial temperature gradients relative to instantaneous spatial mean $\tilde{\mathbf{T}}_{grad}=\mathbf{T}-(\frac{1}{N}\mathbf{1}_N^T\mathbf{T})\mathbf{1}_N$, and $\lambda=200.0$ penalizes the decoupled non-negative spatio-temporal thermal constraint slack variables $s_{i,j}\ge 0$. Because $s_{i,j}\ge 0$ relaxes Eq. (27) to guarantee recursive QP feasibility, $T_{cap}=83.0$°C acts as a *penalized soft thermal cap* rather than a hard barrier, permitting small, bounded transient excursions above 83.0°C while maintaining a 2.0 K safety buffer below the adopted 85.0°C upper simulation thermal limit.

Solver Convergence and Explicit Non-Convergent Fallback Policy: The condensed quadratic program is solved using a warm-started ADMM-style splitting method (Boyd et al., 2011; Stellato et al., 2020) with row-equilibrated constraint matrix $\mathbf{A}$ (reducing $\kappa_2(\mathbf{A})$ from 207.1 to 9.64), over-relaxation parameter $\alpha=1.6$, step size $\rho_{\mathrm{ADMM}}=1.0$, regularization $\sigma=10^{-6}$, and a fixed cap of 60 iterations. OSQP-style scaled primal and dual convergence tolerances ($\varepsilon_{pri},\varepsilon_{dua}$) and formal diagnostic statuses (`converged`, `max_iter`, `soft_constraint_violation`, `numerical_failure`) are evaluated at every update. Whenever an ADMM call does not satisfy both $r_{pri}\le\varepsilon_{pri}$ and $r_{dua}\le\varepsilon_{dua}$ within 60 iterations (`converged = False`), the unconverged iterate is discarded and an explicit deterministic fallback policy ($\mathbf{f}_{\mathrm{fb}}(k)$) is triggered: (i) the controller retrieves the shifted control tail $\mathbf{F}_{\mathrm{conv}}[\min(k_{\mathrm{since}},H_c-1)]$ from the last converged trajectory (or $\mathbf{f}(k-1)$), clamped to $[f_{min},\min(f_{max},\mathbf{f}_{nom}(k))]$; (ii) if any estimated die temperature approaches the soft cap ($\hat{T}_i\ge T_{cap}-0.5$ K), a smooth preventive derating factor $(T_{bar}-\hat{T}_i)/(T_{bar}-(T_{cap}-2.0))$ is applied; and (iii) if total power at the candidate frequency exceeds $P_{bud}(k)$, the frequencies are analytically scaled toward f_{min} via $(P_{bud}(k)-P_{min,cores})/(P_{\mathrm{cand}}-P_{min,cores})$ to guarantee strict compliance with the physical power budget. Finally, a reactive Deterministic Safety Override (DSO) defense-in-depth heuristic barrier operates at the plant rate ($\Delta t_{\mathrm{plant}}=0.5$ s), intervening if estimated temperatures exceed $T_{bar}=84.5$°C to clamp frequencies according to $f_i\le f_{nom,i}\max[0.15,(85.0-T_i)/(85.0-75.0)]$.

LTV Prediction Fidelity Across the Horizon: Because the LTV predictor freezes the local electrothermal Jacobian ($\partial P_{leak}/\partial T_i$, $\partial P/\partial f_i$) and radiator slope at step k across the 30-step (60.0 s) open-loop horizon, non-linear leakage feedback and T^4 radiation introduce growing open-loop prediction drift at long lookahead times. Independent open-loop validation against the non-linear RK2 plant across three operating regimes reveals that: (i) in nominal steady operation, the maximum prediction error across dies is 0.042 K at 2.0 s ($j=1$), 0.229 K at 10.0 s ($j=5$), 0.650 K at 30.0 s ($j=15$), and 1.632 K at 60.0 s ($j=30$, horizon RMSE = 0.802 K); and (ii) under active DVFS heating, the open-loop error is 0.141 K at 2.0 s, 0.613 K at 10.0 s, 2.601 K at 30.0 s, and reaches 4.561 K at the end of the 60.0 s horizon (horizon RMSE = 2.398 K). In closed-loop receding-horizon operation, this 60 s open-loop tail drift does not induce violations of the 85.0°C limit because only the first control move ($j=1$, $\Delta t_{\mathrm{MPC}}=2.0$ s, where

Table 2
System Simulation, Electrothermal, and Orbital Parameters with Explicit Provenance Classification

Category	Symbol & Physical Meaning	Nominal Value	Provenance Class
Lateral Resistances	$R_{12} = R_{34}$, $R_{13} = R_{24}$, $R_{14} = R_{23}$ (inter-node carrier conduction)	1.45, 1.52, 3.20 K/W	Adopted modeling assumption[a]
Die-to-Radiator Res.	R_r (conductive resistance from module node to radiator core)	2.10 K/W	Adopted modeling assumption[a]
Node Capacitances	c_i (effective local module node: die + Cu/AlSiC lid + TIM + carrier footprint)	18.5 J/K	Calculated/assumed composite[b]
Radiator Capacitance	$C_{rad} = m_{rad} c_{p,rad}$ (1.45 kg Al 6061-T6, c_p = 896 J/(kg·K))	1,299.2 J/K	Calculated from geometry/mass
Radiator Optical Prop.	A_{rad} (panel area), ε (IR emissivity), α_s (solar absorptivity)	0.12 m^2, 0.88, 0.12	Literature (white coating) (Pederson et al., 2025)
Orbital Environment	h, T_{orb} (from R_E, μ_E), view factor F, Earth albedo a_E, β	500 km, 5,668.14 s, 0.42, 0.30, 0°	Analytical / Literature (Gilmore, 2002)
DVFS Domain	V_{min}–V_{max}, f_{min}–f_{max} (nominal CMOS operating range)	0.65–1.05 V, 0.30–1.30 GHz	Adopted modeling assumption[c]
Power & Leakage	C_{eff} (switching cap.), P_{ref} (T_0 = 300 K, V_0 = 0.85 V), E_a	24.5 nF, 0.80 W, 0.20 eV	Adopted modeling assumption[c]
State Observer (EKF)	Sensor noise σ_s, process/measurement covariances $\mathbf{Q}, \mathbf{R}$	σ_s = 0.25 K, diag$(0.02\mathbf{I}_4, 0.05)$	Design tuning (Zhang and Srivastava, 2011)
MPC & Safety Limits	Δt_{MPC}, H_c, H_p, soft cap T_{cap}, DSO T_{bar}, sim. limit T_{lim}	2.0 s, 8, 30, 83.0, 84.5, 85.0 °C	Controller design parameters
Workload Proxies	Nominal inference rates (FPS$_{nom}$ at f_{max})	45, 60, 30, 80 FPS	Synthetic benchmark profile

[a] Synthetic lumped-network resistances scaled to represent order-of-magnitude conductive coupling across a shared metallic/ceramic MCM carrier (Wang et al., 2026; European Cooperation for Space Standardization, 2008). [b] Represents the lumped local thermal mass of each SoM subassembly (silicon die < 0.2 J/K plus local lid/heat spreader and substrate region), not bare silicon alone. [c] Synthetic electrothermal parameters scaled to produce 10–25 W per-module dissipation consistent with embedded accelerator literature (Reuther et al., 2022; Kang and Park, 2025); not calibrated to a specific commercial datasheet.

prediction error is bounded by 0.04–0.31 K) is applied before the EKF updates the state and the LTV matrices are re-linearized at $t + 2.0$ s, while the 2.0 K buffer between $T_{cap} = 83.0$°C and 85.0°C absorbs short-term linearization residuals.

5. Computational Simulation Methodology

The multi-physics co-simulation platform was developed in Python 3.14 using NumPy, following verification and credibility principles for computational modeling and simulation (National Aeronautics and Space Administration, 2026). The numerical simulation contains 34,009 time samples and 34,008 integration intervals of $\Delta t_{\text{plant}} = 0.5$ s integrated via a 2nd-order Runge-Kutta scheme (Heun / RK2) with trapezoidal power integration, spanning 17,004.0 s, compared with an analytical three-orbit duration of approximately 17,004.43 s ($3 \times 5,668.14$ s). A formal temporal convergence study across $\Delta t_{\text{plant}} \in \{1.0, 0.5, 0.25, 0.125\}$ s (with strictly invariant control step $\Delta t_{\text{MPC}} = 2.0$ s and scheduling period $\Delta t_{sched} = 120.0$ s) confirmed monotonic convergence: the nominal discretization ($\Delta t = 0.5$ s) incurs an absolute deviation of only 0.034 K in peak junction temperature ($T_{\max} = 83.211$°C vs. 83.177°C at $\Delta t = 0.125$ s), a peak spatial gradient difference of 0.047 K, and a relative total energy difference of 0.0114%, remaining well below the sensor measurement noise floor ($\sigma_s = 0.25$ K).

Benchmarking Protocol: To ensure strict methodological parity, all benchmarked governors evaluate:

(i) identical initial states $\mathbf{x}_0$ ($T_{i,0} = 310$ K, $T_{rad,0} = 270$ K);
(ii) identical synthetic inference workload request traces $\mathbf{f}_{nom}(t)$ and activity factors $\alpha(t)$;
(iii) identical modeled orbital environmental radiative forcing $Q_{ext}(t)$;
(iv) identical stochastic sensor noise realizations for any given seed;
(v) identical thermal plant model and integration steps.

Performance and Energy Proxies: Aggregate inference throughput is evaluated via the performance proxy:

$$\text{FPS} = \frac{1}{N_{steps}} \sum_t \sum_{i=1}^{N} \frac{f_i(t)}{f_{max}} \text{FPS}_{nom,i}, \tag{29}$$

and aggregate energy efficiency is quantified by the throughput-normalized energy-delay product proxy:

$$E_{t_{\text{frame}}} = E \cdot \frac{1}{\text{FPS}} \quad [\text{kJ} \cdot \text{s/frame}], \tag{30}$$

where $E = \Delta t_{\text{plant}} \sum_t \sum_i P_i(t)$. Instantaneous inter-die spatial gradient is computed as $\Delta T_{esp}(t) = \max_i T_i(t) - \min_i T_i(t)$.

Benchmarked Control Policies:

Table 3
Consolidated Results in 3-Orbit LEO Co-Simulation (500 km SSO, 34,009 Steps, $\Delta t = 0.5$ s, Seed 0)

Strategy	T_{mean} (°C)	$T_{\max}$ (°C)	ΔT_{peak} (K)	ΔT_{mean} (K)	$t_{>85}$ (s)	FPS	$E_{t_{\text{fr}}}$ (kJ·s/fr)	Energy (kJ)
Linux Ondemand	350.17†	412.30†	12.87	8.64	16,038.0	181.21†	28.38	5,143.3†
Ondemand + trip 95/85	87.96	95.95	9.28	3.93	16,037.5	117.27	8.63	1,011.6
Throttling 85/75	78.08	86.06	9.42	5.26	200.5	119.58	7.57	905.7
Throttling 83/73	76.17	83.78	9.46	5.24	0.0	118.75	7.46	885.4
Space-PID	79.73	85.56	4.88	0.62	14.0	123.24	7.47	920.5
Power Cap (65 W)	93.95	109.31	5.82	3.66	14,983.5	126.49	8.55	1,081.3
TCA-DVFS (simple)	80.53	83.68	1.58	0.45	0.0	116.16	8.03	932.6
TCAS-DVFS (Proposed)	**80.15**	**83.25**	**2.01**	**0.89**	**0.0**	**124.62**	**7.45**	**928.6**

† Regimes beyond model validity ceiling (> 125°C). Linux Ondemand experiences thermal runaway under positive subthreshold leakage feedback, reaching 412.30 °C and dissipating an unphysical average of 302.5 W. This regime represents numerical extrapolation failure and is strictly excluded from quantitative reliability and performance inferences. Overshoot to 86.06 °C in Throttling 85/75 stems from sampling latency and sensor noise.

1. *Linux Ondemand:* standard reactive governor without thermal protection.
2. *Ondemand + trip 95/85:* ondemand governor with emergency trip hysteresis at 95 °C.
3. *Thermal Throttling 85/75:* binary frequency drop to f_{min} upon reaching 85 °C with recovery at 75 °C.
4. *Thermal Throttling 83/73:* conservative baseline matched to the 83 °C design margin.
5. *Space-PID:* decentralized per-die PID controllers with gains selected from a predefined 27-point grid by minimizing an ITAE-based objective augmented with a thermal-violation penalty over the initial tuning interval ($K_p = -0.07$, $K_i = -0.002$, $K_d = -0.04$) set to regulate at 80 °C.
6. *Power Cap 65 W:* static global power budgeting enforced by the EPS.
7. *TCA-DVFS (simple):* uncoordinated Laplacian rule with thermal barrier and radiative budget, excluding MPC and SDP.
8. **TCAS-DVFS (proposed):** full multi-rate architecture integrating EKF, topological SDP task allocation, and warm-started ADMM-based discrete LTV MPC-DVFS.

All simulation source code, experimental pipelines, nominal configuration manifests (`configs/nominal.json`), convergence verification suites, ASTM E1049-85 validation test harnesses, and computational reproducibility metadata (`results/metadata.json`) are organized under an open-source MIT repository structure at https://github.com/dafnhe-thor-barros/TCAS-DVFS.

6. Results and Discussion

6.1. Global Performance Comparison

Table 3 presents the consolidated comparative results across 3 full orbits. **TCAS-DVFS** successfully maintained junction temperatures strictly below the adopted 85 °C simulation thermal limit, registering a peak of 83.25°C and **zero observed seconds of thermal violation** (0.00%). Because $T_{cap} = 83.0$°C is formulated as a soft thermal cap with slack variables $s_{i,j} \geq 0$, the controller permits a bounded 0.25 K excursion above 83.0°C during peak solar loading while remaining 1.75 K below the 85.0°C limit. In contrast, standard Thermal Throttling 85/75 accumulated 200.5 s of overheating due to hysteresis latency, and Space-PID breached the limit for 14.0 s during the post-eclipse thermal shock.

Across the 8,502 online MPC updates executed during the 3-orbit nominal campaign, the warm-started ADMM solver satisfied the scaled primal-dual stopping criteria ($\varepsilon_{pri}, \varepsilon_{dua}$) within the 60-iteration budget in 94.31% of calls ($8,018/8,502$, averaging 29.7 iterations per update), with zero numerical failures (`NaN`/`Inf`) and zero physical power-budget infeasibilities ($P_{bud}(k) \geq P_{min,cores}(k)$ at all steps). In the remaining 484 updates (5.69%), which occurred during rapid workload-step transitions where the dual residual r_{dua} remained marginally above ε_{dua} at iteration 60 ($r_{pri} \leq 1.76 \times 10^{-2}$, $r_{dua} \leq 6.42 \times 10^{-2}$), the explicit non-convergent fallback policy ($\mathbf{f}_{\text{fb}}$) was triggered 484 times. During all fallback activations, physical junction temperatures remained strictly bounded below 83.25°C (with noisy EKF state estimates reaching at most $T_{\max,\text{fb}} = 83.42$°C), confirming that the deterministic power-projected fallback prevents any control degradation or thermal overshoot when the iterative solver reaches its fixed iteration ceiling.

In terms of processing throughput, TCAS-DVFS delivered 124.62 FPS, outperforming Throttling 85/75 (+4.2%) and conservative Throttling 83/73 (+4.9%), while achieving the lowest energy-delay product proxy among all evaluated policies (7.452 kJ · s/frame versus 7.456 kJ · s/frame for Throttling 83/73 and 7.574 kJ · s/frame for Throttling 85/75). Relative to simple uncoordinated TCA-DVFS (116.16 FPS, 8.028 kJ · s/frame), the constrained MPC controller delivered a +7.3% throughput boost and a 7.2% reduction in $E_{t_{\text{frame}}}$, demonstrating the practical efficacy of active multivariable coordination.

Table 4 details performance across 10 independent stochastic noise seeds. EKF filtering maintains maximum temperature dispersion to within ± 0.02 K in TCAS-DVFS, compared to ± 0.15 K for Throttling.

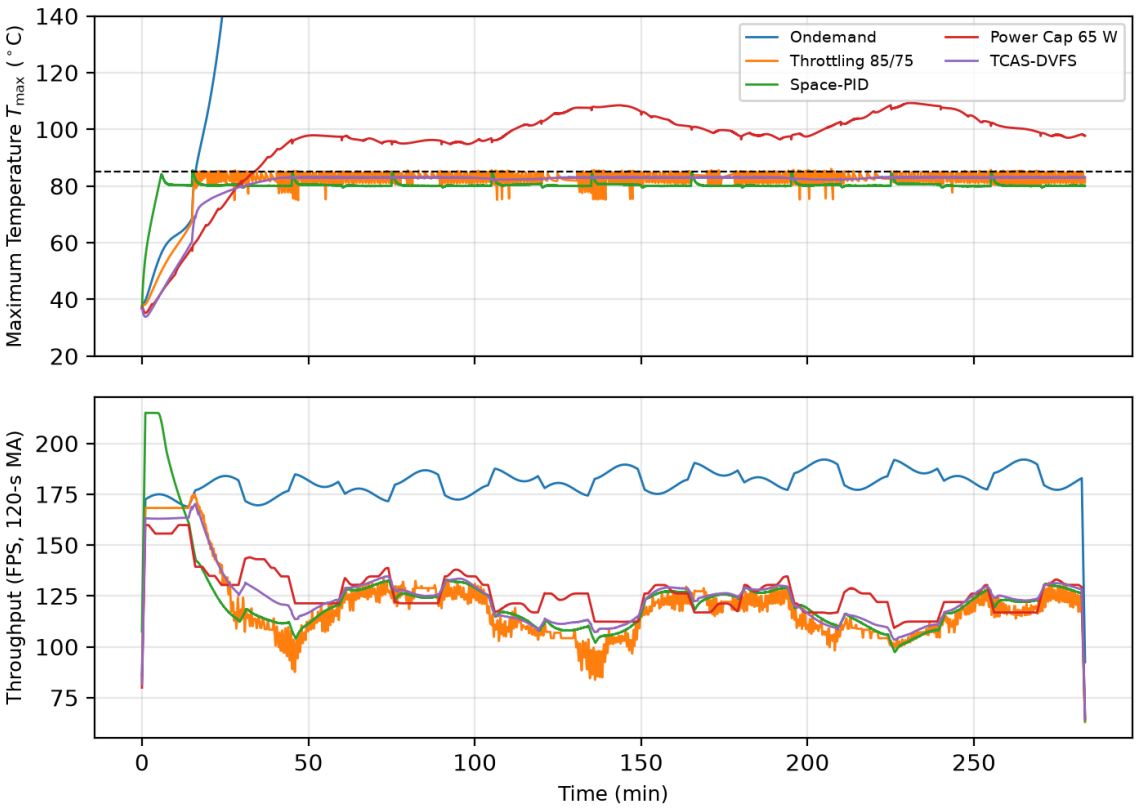


Figure 4: Temporal co-simulation dynamics over 3 complete orbits (seed 0): (a) maximum silicon junction temperature ($T_{\max}$), demonstrating strict compliance of TCAS-DVFS below the 85 °C simulation limit versus saw-tooth hysteresis in Throttling; (b) aggregate AI inference throughput smoothed with a 120 s moving average.

Table 4
Statistical Dispersions across 10 Stochastic Sensor Noise Seeds (Seeds 0 to 9)

Metric	Throttling 85/75	Space-PID	TCAS-DVFS
$T_{\max}$ (°C)	85.89 ± 0.15	85.54 ± 0.04	$\mathbf{83.24 \pm 0.02}$
$t_{>85}$ (s)	213.5 ± 15.7	14.6 ± 0.7	$\mathbf{0.0 \pm 0.0}$
ΔT_{peak} (K)	9.48 ± 0.09	4.86 ± 0.02	$\mathbf{2.00 \pm 0.01}$
ΔT_{mean} (K)	5.26 ± 0.13	0.617 ± 0.001	$\mathbf{0.886 \pm 0.001}$
FPS	119.70 ± 0.35	123.24 ± 0.00	$\mathbf{124.62 \pm 0.00}$
$E_{t_{\text{frame}}}$ (kJ·s/frame)	7.566 ± 0.024	7.469 ± 0.000	$\mathbf{7.452 \pm 0.000}$
Energy (kJ)	905.7 ± 0.3	920.47 ± 0.01	$\mathbf{928.64 \pm 0.01}$

6.2. Thermal Dynamics and Gradient Suppression

Figs. 4, 5, and 6 illustrate the orbital transient responses. While Throttling induces saw-tooth oscillations near 10 K and Space-PID exhibits gradient excursions during daylight transitions ($\Delta T_{\text{peak}} = 4.88$ K), TCAS-DVFS dampens mean gradients to 0.89 K (an 83.1% reduction) and peak gradients to 2.01 K (a 78.7% reduction relative to Throttling). Operating temperatures remained above the nominal 83 °C soft cap for 44.5% of the three-orbit simulation duration (7,571.0 s out of 17,004.0 s), confirming that the predictive controller deliberately tracks the upper thermal boundary to maximize compute throughput while never activating the 84.5°C safety override ($n_{bar} = 0$).

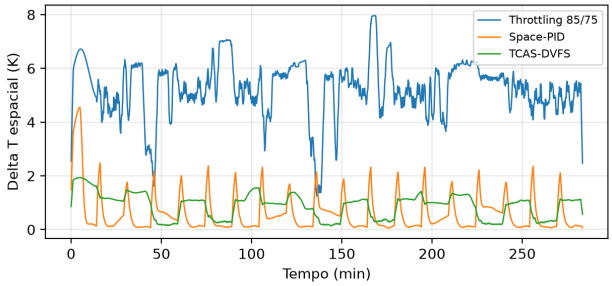


Figure 5: Temporal evolution of inter-die spatial thermal gradient ($\Delta T_{esp}(t) = \max_i T_i - \min_i T_i$) with a 120 s moving average: active suppression of conductive crosstalk by TCAS-DVFS.

6.3. Functional Ablation Study

The ablation results in Table 5 (evaluated over 1 orbit, 11,337 steps) isolate the architectural mechanisms:

- **Dominant observed contribution of MPC-DVFS in the tested configuration:** removing the MPC controller (*Without MPC*) immediately leads to thermal threshold exceedance, accumulating 463.5 s above 85 °C, a peak of 85.41°C, and an increase in peak

Table 5
Functional Ablation Study of TCAS-DVFS (1 Orbit, Seed 0)

Variant	$T_{\max}$ (°C)	ΔT_{peak} (K)	ΔT_{mean} (K)	FPS	$E_{t_{\text{frame}}}$ (kJ·s/frame)	$t_{>85}$ (s)
Full ($H_p = 30$, 60.0 s)	83.21	2.01	1.00	132.92	2.504	0.0
Without SDP	83.21	2.01	0.98	133.25	2.499	0.0
Without MPC	85.41	7.09	1.59	123.11	2.801	463.5
$H_p = 10$ (20.0 s)	83.21	2.00	1.00	132.91	2.504	0.0
$H_p = 60$ (120.0 s)	83.21	2.42	1.00	132.91	2.504	0.0
Simple TCA	83.66	1.58	0.60	126.28	2.645	0.0

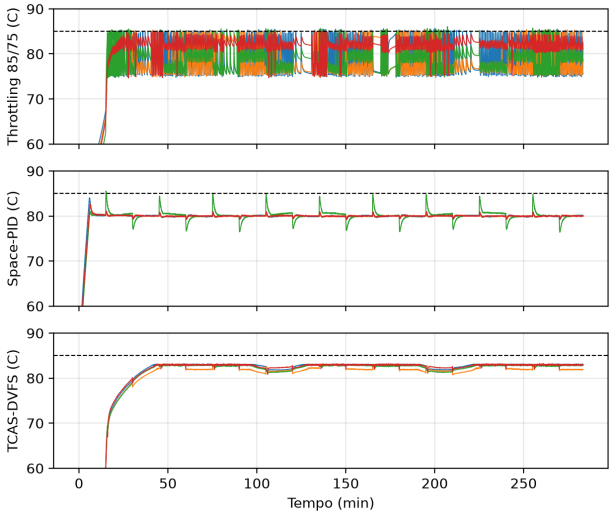


Figure 6: Individual junction temperature profiles for the 4 silicon dies (seed 0): homogeneous convergence under coordinated TCAS-DVFS versus spatial divergence under classical baselines.

gradient to 7.09 K despite 8,573 activations of the reactive safety backstop. Dynamic lookahead within the MPC is the primary driver of proactive containment.

- **Limited observed benefit of SDP scheduling for the tested $N = 4$ configuration:** the *Without SDP* variant (static task assignment) achieves performance essentially identical to the full framework (83.21°C, 133.25 FPS, 2.499 kJ · s/frame). Although randomized hyperplane rounding over the rank-regularized SDP relaxation matched the best discrete permutation found by exhaustive search in 100.0% of calls (48/48 calls in 1 orbit and 142/142 calls in 3 orbits, performing 12 and 36 task migrations, respectively, despite ADMM numerical convergence occurring in only 42.0% of warm-started benchmark calls), for $N = 4$ dies there are only $4! = 24$ permutations and conductive cross-coupling is spatially balanced. Consequently, topological remapping is largely neutral once the migration stall penalty is accounted for, and its contribution in compact four-die clusters is secondary to predictive DVFS.

- **No material difference observed at the reported precision for the tested horizons:** varying prediction horizon $H_p \in \{10, 30, 60\}$ steps (corresponding to physical lookahead intervals of 20.0, 60.0, and 120.0 s under $\Delta t_{\text{MPC}} = 2.0$ s) produced nearly identical performance metrics (132.91–132.92 FPS, 2.504 kJ · s/frame). Crucially, reducing the horizon to $H_p = 10$ (20.0 s) achieves 100.0% ADMM convergence within 60 iterations (2,834/2,834 calls, averaging 20.1 iterations with zero fallback activations) while bounding open-loop LTV linearization drift below 1.5 K, confirming that the 4.56 K open-loop prediction error at 60.0 s does not degrade closed-loop regulation.

6.4. Parametric Sensitivity Analysis (Stratified Latin Hypercube Sampling, $n = 30$)

To evaluate descriptive sensitivity to variations in the passive conductive and capacitive thermal network, a 30-sample stratified Latin Hypercube Sampling (LHS) campaign was conducted (3 full orbits per sample, evaluating 1,020,270 integration steps) with simultaneous continuous uniform perturbations of ±20% across the four network parameters R_{ij}, R_r, c_i, and C_{rad}, while holding radiator optical properties (ε, α_s), radiator area (A_{rad}), electrical power parameters, and external orbital fluxes at their nominal values.

Because LHS is a stratified sampling design without replacement rather than an independent and identically distributed (IID) Bernoulli sequence, the campaign is reported strictly as a descriptive sensitivity analysis across the sampled four-parameter space without applying binomial Clopper-Pearson confidence bounds. No thermal-threshold violations above the 85°C simulation limit were observed across the 30 sampled realizations (0/30 samples > 85°C, with mean $T_{\max} = 82.35 \pm 1.58$°C, 95% CI: $[81.76, 82.94]$°C, bootstrap 95% interval: $[81.77, 82.89]$°C), whereas Throttling 85/75 exceeded 85°C in all 30/30 samples (mean violation time 234.8 s). Across all 255,060 MPC updates executed in the 30 LHS trajectories under plant-model mismatch, the ADMM solver converged within 60 iterations in 84.76% of calls (216,178/255,060) and triggered the explicit safe fallback policy in the remaining 15.24% (38,882 calls, with peak EKF temperature during fallback of 83.83°C) without any violation of the 85°C limit.

In paired comparisons under identical physical parameters:

(i) *Versus Throttling 83/73*: TCAS-DVFS reduced peak temperatures by 1.51 K (95% CI: $[-2.10, -0.93]$ K), attenuated mean and peak spatial gradients by -82.48% and -78.46%, improved inference throughput by $+4.95\%$ (95% CI: $[+4.11\%, +5.80\%]$), and reduced the EDP proxy by -1.23% (95% CI: $[-1.79\%, -0.67\%]$).

(ii) *Versus Throttling 85/75*: TCAS-DVFS reduced peak temperature by 3.54 K (95% CI: $[-4.12, -2.96]$ K), attenuated mean and peak spatial gradients by -82.55% and -78.58%, raised throughput by $+4.08\%$ (95% CI: $[+3.22\%, +4.93\%]$), and achieved an EDP reduction of -2.63% (95% CI: $[-3.20\%, -2.06\%]$).

Within the sampled four-parameter uncertainty space, R_r exhibited the strongest monotonic association with $T_{\max}$, with Spearman rank correlation $\rho = +0.995$, followed by die heat capacity c_{die} ($\rho = -0.338$), radiator capacitance C_{rad} ($\rho = -0.234$), and inter-die crosstalk resistance R_{ij} ($\rho = -0.173$).

7. Indicative Reliability and Physical Degradation Assessment

All reliability projections utilize thermal histories after discarding the initial 2-hour transient warm-up phase, and are presented strictly as dimensionless comparative damage indices ($D_{\text{base}}/D_{\text{prop}}$) rather than absolute lifetime (MTTF) predictions in hours.

7.1. Electromigration (Black's Equation Damage Proxy)

A relative Black-equation damage-rate proxy was computed from the simulated electrical-power and junction-temperature histories (Black, 1969):

$$\text{MTTF}_{EM} \propto J^{-n} \exp\left(\frac{E_{a,EM}}{k_B T_j}\right), \tag{31}$$

where $n = 1.8$, $E_{a,EM} = 0.70$ eV, and current density is approximated by the normalized electrical current proxy $J \propto P/V$ under uniform effective interconnect cross-section. Time-integral accumulation over 3 orbits yielded relative damage ratios ($D_{\text{ref}}/D_{\text{prop}}$) of **0.97** (aggregate dies) and 1.42 (worst die) versus Throttling 85/75; and **0.83** and 1.23 versus Throttling 83/73. Because TCAS-DVFS draws $+7.6\%$ and $+9.9\%$ higher average current due to sustained workload execution, we formally conclude that **the framework provides no net electromigration life extension**.

7.2. Thermomechanical Solder Fatigue (Coffin-Manson Cyclic Severity)

Cumulative low-cycle thermal cycling severity in SAC305 solder interconnects is estimated using the Coffin-Manson relation (Manson, 1966):

$$N_f = B\,(\Delta T_{cycle})^{-q}, \quad q = 2.1. \tag{32}$$

Table 6
Rainflow Cycle Counting and Relative Fatigue Damage Severity Ratios ($q = 2.1$, Aggregate Dies)

Threshold (ΔT)	Cycles Throttling	Cycles TCAS-DVFS	Damage Ratio ($D_{\text{thr}}/D_{\text{prop}}$)
0.25 K	741	259	680.7
1.00 K	724	29	1,127.8
2.00 K	612	8	3,171.7

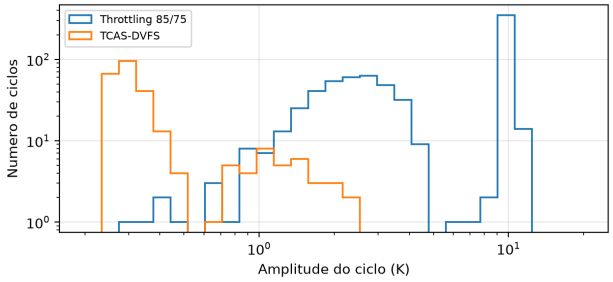


Figure 7: Rainflow thermal cycle range spectrum (ΔT, 0.25 K threshold after warm-up): TCAS-DVFS suppresses nearly all damaging cycles ($\Delta T > 2$ K).

Applying a three-point rainflow algorithm formally validated against the standard test sequences of ASTM E1049-85 (Section 5.4.4) and synthetic benchmarks to confirm exact peak-to-peak range counting ($\Delta T = T_{\text{peak}} - T_{\text{valley}}$), the maximum cyclic thermal excursion after transient warm-up dropped from 11.20 K (Throttling) to 2.39 K (TCAS-DVFS).

As shown in Table 6 and Fig. 7, eliminating large thermal excursions ($\Delta T > 2$ K) noticeably slows theoretical cyclic fatigue accumulation. However, junction temperature serves strictly as an indicative proxy for thermal cycling amplitude severity, rather than viscoplastic shear strain in actual solder joints. Comprehensive finite-element thermo-mechanical analysis remains required for flight qualification.

8. Related Work

Dynamic Thermal Management (DTM) in terrestrial multi-core architectures has been extensively investigated using tools such as HotSpot (Skadron et al., 2004; Brooks and Martonosi, 2001). 3D floorplanning (Lin et al., 2021), distributed task migration (Ge et al., 2010), and convex frequency optimization (Murali et al., 2007) advanced thermal regulation under convective conditions. In the aerospace domain, autonomous execution was validated by Sherwood et al. (2003), while advanced packaging models were proposed by Wang et al. (2026), and GPU orbital thermal behavior was characterized by Yuan et al. (2025). The closest prior work is by Chen et al. (2026b), which formulated thermal crosstalk scheduling. Table 7 positions our contribution relative to the state of the art.

Table 7
Positioning against Closely Related Prior Work

Work	Sched.	Pred. DVFS	EKF	T^4 + Ecl.
Murali et al. (2007)	No	Static	No	No
Yuan et al. (2025)	No	No	No	Empirical
Chen et al. (2026b)	Yes	No	No	No
This Work	**Yes**	**LTV MPC**	**Yes**	**Yes**

9. Practical Implementation Considerations and Threats to Validity

9.1. Implementation Constraints

Deploying TCAS-DVFS on spaceborne flight computers requires addressing online execution constraints for future real-time-oriented embedded deployment:

1. **Execution Complexity:** Embedded execution time, memory footprint, and worst-case timing remain to be benchmarked on flight-representative hardware. The warm-started ADMM-style iteration operates with a fixed cap of 60 iterations, early exit monitoring, and an explicit deterministic shifted-tail fallback policy to guarantee bounded execution time and safe actuation when the iteration limit is reached.
2. **Memory Footprint:** The complete EKF covariance matrices, discrete prediction models, and warm-start vectors require modest scratchpad memory fitting within typical rad-hard microcontrollers.

9.2. Threats to Validity

As a model-based numerical investigation, several limitations must be explicitly recognized:

1. **Absence of Physical and Environmental Testing:** The findings rely entirely on numerical simulation of the stated lumped-parameter thermal network and synthetic analytical orbital forcing, and have not been validated via Hardware-in-the-Loop (HIL) testbeds or Thermal Vacuum (TVAC) chamber testing.
2. **Lumped Modeling and Linearization Drift:** While the 5×5 CTM captures inter-die conductive coupling and bulk radiator dynamics, it does not resolve spatial micro-gradients across silicon dies or three-dimensional heat spreader diffusion. Furthermore, open-loop LTV linearization exhibits up to 4.56 K prediction drift at a 60 s horizon under active heating, relying on 2.0 s receding-horizon relinearization and the 2.0 K soft-cap margin for closed-loop containment.
3. **Throughput and Reliability Proxies:** Workload throughput is evaluated via a linear frequency proxy rather than cycle-accurate kernel profiling. Similarly, Black's equation and Coffin-Manson models are indicative proxies based on temperature histories rather than direct measurements of current density or plastic strain.
4. **Omission of Space Radiation Effects:** The present study focuses strictly on electrothermal dynamics and does not model Single Event Effects (SEE) or Total Ionizing Dose (TID), which can affect DVFS stability and memory integrity in orbital environments.

10. Conclusions

This paper presented TCAS-DVFS, a predictive control framework designed to mitigate spatial thermal crosstalk in high-density spaceborne AI clusters operating in deep orbital vacuum. The conclusions of this model-based numerical study are delineated into three levels:

1. **Demonstrated in the nominal model:** Online EKF state estimation, coupled radiative-conductive power budgeting, and discrete LTV MPC-DVFS (with a penalized 83 °C soft thermal cap and explicit non-convergent fallback) produced zero observed thermal violations above the adopted 85 °C simulation thermal limit over 3 complete simulated LEO orbits (17,004.0 s), attenuated peak and mean inter-die spatial gradients by 78.7% and 83.1%, respectively, and sustained 124.62 FPS. Ablation indicates a dominant contribution from the predictive control formulation in damping gradients and bounding temperatures, whereas SDP allocation matches exhaustive permutation search across the $4! = 24$ discrete candidates after rounding (100% hit rate despite 42% warm-start ADMM convergence) but yields negligible macroscopic thermal benefit in compact symmetric $N = 4$ arrangements.
2. **Observed under descriptive parametric sensitivity sampling:** Under a $\pm 20\%$ stratified Latin Hypercube Sampling campaign ($n = 30$) perturbing the four conductive and capacitive network parameters ($R_{ij}, R_r, c_i, C_{rad}$), zero violations above 85 °C were observed ($0/30$ samples), with sensitivity heavily dominated by die-to-radiator thermal resistance (R_r, Spearman $\rho = +0.995$). Because LHS is a non-IID stratified design, these results represent descriptive sensitivity within the sampled four-parameter envelope rather than a probabilistic reliability certification.
3. **Unverified physically:** The reported evidence is limited to numerical simulation. Physical service life, flight qualification, TVAC chamber behavior, embedded execution timing, and radiation tolerance remain unverified and constitute mandatory steps for future experimental development.

CRediT Authorship Contribution Statement

Dafnhê T. C. M. Barros: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft. **Flávio B. Braz da Silva:** Supervision, Project administration, Resources, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used Anthropic Claude and Google Gemini in order to assist with English language stylistic editing, simulation script refactoring, and conceptual visual rendering of orbital geometry and thermal crosstalk (Figs. 1 and 2). After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability

All simulation source code, execution pipelines, nominal parameter manifests, co-simulation logs, and computational reproducibility metadata are openly available in the GitHub repository at https://github.com/dafnhe-thor-barros/TCAS-DVFS.

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